

Supplements

1 Low channel

Table 1 BCCS-NNLS fitting results for mixed sources using BCCS subtracted method. 3-5 columns are contributions obtained using the full 256 channels and 6-8 are contributions obtained using the low 150 channels

Samples	Distance (cm)	Full channel NNLS			Low channel NNLS		
		Am-241	Co-60	Cs-137	Am-241	Co-60	Cs-137
Co-60/Cs-137	136/136	0.1117	0.2856	0.3707	0.1116	0.2856	0.3707
Am-241/Co-60	287/287	0.1436	0.0688	0.0080	0.1436	0.0688	0.0080
Co-60/Cs-137	361/361	0.0753	0.0461	0.0689	0.0752	0.0461	0.0689
Am-241/Cs-137	163/163	0.4090	0.0056	0.2600	0.4089	0.0056	0.2600
Am-241/Co-60/Cs-137	309/309/309	0.1909	0.0673	0.0876	0.1909	0.0673	0.0876
Am-241/Co-60	210/153	0.2523	0.2307	0.0100	0.2522	0.2306	0.0101
Co-60/Cs-137	285/367	0.0728	0.0774	0.0652	0.0728	0.0774	0.0652
Am-241/Cs-137	85/173	0.9920	0.0101	0.2299	0.9920	0.0101	0.2299
Co-60/Cs-137	223/316	0.0954	0.1214	0.0858	0.0953	0.1214	0.0859

Table 2 BCCS-NNLS fitting results for mixed sources using BCCS ratio method. 3-5 columns are contributions obtained using the full 256 channels and 6-8 are contributions obtained using the low 150 channels

Samples	Distance (cm)	Full channel NNLS			Low channel NNLS		
		Am-241	Co-60	Cs-137	Am-241	Co-60	Cs-137
Co-60/Cs-137	136/136	0.9870	0.3302	0.2813	0.1554	0.3094	0.3311
Am-241/Co-60	287/287	0.6267	0.0713	0.0000	0.1996	0.0693	0.0070
Co-60/Cs-137	361/361	0.2036	0.0562	0.0613	0.0844	0.0520	0.0589
Am-241/Cs-137	163/163	0.8446	0.0159	0.2417	0.3683	0.0085	0.2551
Am-241/Co-60/Cs-137	309/309/309	0.1110	0.0772	0.0784	0.2199	0.0746	0.0743
Am-241/Co-60	210/153	0.6394	0.2622	0.0000	0.2468	0.2462	0.0000
Co-60/Cs-137	285/367	0.4550	0.0925	0.0453	0.2458	0.0889	0.0446
Am-241/Cs-137	85/173	0.3696	0.0129	0.2388	0.8101	0.0144	0.2238
Co-60/Cs-137	223/316	0.7239	0.1523	0.0259	0.1876	0.1378	0.0568